FORMATIVE ASSIGNMENT

EduBridge Africa – A Digital Platform for Early Career Discovery and Skills Alignment

- **Description of the concept of their Mission;**

  Sector: ***Education & Youth Development (EdTech).***

  My mission is to empower Rwandan secondary school students through technology by improving access to career discovery and guidance. This aligns with ALU's goal of developing leaders who solve Africa's challenges. Rwanda's Vision 2050 emphasizes building a knowledge-based economy, yet many students lack access to career counseling—the Rwanda Education Board (2023) reports only one counselor per 1,500 students. EduBridge Rwanda addresses this gap by providing a simple, accessible digital tool for career exploration targeted at Senior 1-6 students in Rwandan schools.

- **Problem statement:**
  - ☐ **WHO:** Secondary school students (Senior 1-6) in Rwanda, aged 13-19, particularly in public schools with limited career guidance.
  - ☐ **WHAT:** Students make uninformed subject and career choices due to lack of self-assessment tools and career information, leading to mismatched university programs.
  - ☐ **WHEN:** Critical decisions occur in Senior 3 (O-Level subject selection) and Senior 6 (university choice).
  - ☐ **WHERE**: Rwanda, focusing on schools in Kigali as a pilot.
  - ☐ **WHY**: Shortage of career counselors (1 per 1,500 students), no localized digital tools, limited awareness of career options.
  - ☐ **HOW SIGNIFICANT:** According to NISR (2022), 38% of Rwandan graduates work in fields unrelated to their studies, wasting resources and contributing to youth underemployment.

Due: 25th January 2026, 23:59 PM

- Each student must design a software solution to a real-world problem. Each student shall be expected to present the problem and the suggested solution in a project proposal format.

- EduBridge Rwanda is a simple web application that helps students discover career paths through an interest questionnaire.
  Core Features:
  (1) Simple career interest questionnaire (25-30 questions),
  (2) Career profiles with descriptions, required skills, and education paths,
  (3) Basic matching of interests to 3-5 suggested careers,
  (4) Information on Rwandan universities and TVET programs.

  What Makes This Unique: Localized for Rwanda (Kinyarwanda language option, local career data), simple enough for low-bandwidth areas, free for students, and designed specifically for the Rwandan education system and job market.

- **Software development model**

  Selected Model: Agile (Scrum).
  Why Agile is Relevant:
  (1) Can start with a simple MVP and add features based on student feedback,
  (2) Allows testing with real students early to ensure usefulness,
  (3) Flexible to change requirements as we learn what works.

  Development Steps:
  Sprint 1 (2 weeks): User registration and basic assessment questions.
  Sprint 2 (2 weeks): Career database and matching algorithm.
  Sprint 3 (2 weeks): Results display and career information pages.
  Sprint 4 (2 weeks): Testing with pilot school, bug fixes, improvements.

  Total: 8 weeks to MVP.

- **The hypothesis of their solution**

If Rwandan secondary school students (S3-S6) use a simple, localized career discovery tool, then they will have better awareness of career options aligned with their interests, leading to more informed subject and program choices.

SMART:

- ☐ Specific (targets S3-S6 students in Rwanda),
- ☐ Measurable (track assessments completed, user satisfaction),
- ☐ Achievable (simple web app with standard technologies),
- ☐ Realistic (pilot with 1-2 schools, 100-200 students),
- ☐ Time-bound (MVP in 8 weeks, pilot results in 12 weeks).

- **References**

Use APA Referencing Style

National Institute of Statistics of Rwanda. (2022). Labour Force Survey Annual Report. NISR. | Rwanda Education Board. (2023). Annual Report on School Counseling Services. REB. | Ministry of Education Rwanda. (2021). Education Sector Strategic Plan 2018-2024. MINEDUC.

The deadline for the submission of the assignment is in the 4th week on 25 January 2026, 23:59 PM.

Software Requirements Specification (SRS) Template

Title: EduBridge Rwanda

Prepared by: Gisele Mwizera Amen

African Leadership University (ALU)

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Revision History

Name	Date	Reason For Changes	Version
Week 4 assignment	24th/01/2026	Initial draft	v1.0

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed description of the requirements for EduBridge Rwanda, a web-based career discovery platform designed specifically for Rwandan secondary school students.

This document covers Version 1.0 (Minimum Viable Product) of the system, which includes the following core functionality:

- User registration and authentication for students
- Career interest assessment questionnaire (25-30 questions)
- Career matching algorithm and results display
- Career information database with Rwandan education pathways
- Bilingual support (English and Kinyarwanda)

This SRS will serve as a contract between the development team and stakeholders, ensuring all parties have a clear understanding of what the system will deliver.

1.2 Document Conventions

Requirements are labeled FR (Functional Requirement) and NFR (Non-Functional Requirement). Priority is implicit based on numbering, with lower numbers indicating higher priority.

1.3 Intended Audience and Reading Suggestions

This document is intended for developers implementing the system, professors, project managers overseeing timelines, and testers validating functionality. Readers should start with the Introduction, then proceed to Chapter 4 (Functional Requirements) and Chapter 5 (Non-Functional Requirements).

1.4 Product Scope

EduBridge Rwanda is a web-based career discovery platform designed to help Rwandan secondary school students (Senior 1-6) explore and identify career paths that align with their interests, skills, and aspirations.

The system WILL:

- Allow students to create accounts and securely log in
- Provide a career interest assessment questionnaire
- Match student interests to relevant careers (top 3-5 matches)
- Display detailed career information (description, skills, education paths, salary ranges)
- Show relevant Rwandan universities and TVET institutions for each career
- Save assessment results for future reference
- Support both English and Kinyarwanda languages

The system will NOT:

- Provide formal career counseling or psychological assessments
- Guarantee career outcomes or job placement
- Require any payment from students (free service)
- Connect students directly with employers or mentors (future feature)
- Provide offline functionality (requires internet connection)

2. Overall Description

2.1 Product Perspective

EduBridge Rwanda is a new, self-contained web application. It is not a replacement for existing systems but rather fills a gap where no localized digital career guidance tool currently exists for Rwandan secondary students.

System Architecture: The system follows a three-tier architecture:

1. Presentation Layer (Frontend): Responsive web interface using HTML5, CSS3, bootstrap
2. Application Layer (Backend): PHP server handling business logic
3. Data Layer (Database): MySQL storing user data, assessments, and career information

Major System Components:

- User Module: Handles registration, authentication, and profile management
- Assessment Module: Manages career interest questionnaire and scoring algorithm
- Results Module: Displays career matches and detailed career/education information

2.2 Product Functions

The major functions of EduBridge Rwanda include:

(1) User Registration and Authentication - students create accounts and securely log in;

(2) Career Interest Assessment - students complete a 25-30 question questionnaire about their interests, skills, and preferences;

(3) Assessment Scoring - the system calculates scores across career categories;

(4) Career Matching - displays top 3-5 career matches with percentage scores;

(5) Career Information Display - shows detailed career profiles including descriptions, required skills, education paths, and salary ranges in Rwanda;

(6) Education Pathway Information - lists relevant Rwandan universities and TVET institutions;

(7) Results Management - saves and allows retaking of assessments;

(8) Language Support - bilingual interface in English and Kinyarwanda.

2.3 User Classes and Characteristics

Primary User Class - Students: Rwandan secondary school students (S1-S6), aged 13-19. They have basic computer literacy, access to smartphones or school computers, and are the main users of the assessment and career exploration features. They require a simple, intuitive interface.

Secondary User Class - School Administrators: Teachers or career guidance staff who may view aggregate data or assist students. They have moderate technical skills and need basic reporting features.

Tertiary User Class - System Administrator: Technical staff responsible for maintaining the system, updating career data, and managing user accounts. They require full system access and technical expertise.

2.4 Operating Environment

The system will operate as a web application accessible via modern web browsers (Chrome, Firefox, Safari, Edge) on desktop computers, laptops, tablets, and smartphones.

The application must be responsive and work on devices with screen sizes from 320px to 1920px.

Server environment: Linux-based cloud hosting (e.g., AWS, DigitalOcean, or local Rwandan hosting).

Database: MySQL or PostgreSQL.

The system must function on 4G mobile networks common in Rwanda and handle intermittent connectivity gracefully.

2.5 Design and Implementation Constraints

The following constraints will limit the development options:

Budget Constraints: The project has limited funding, requiring the use of free or open-source technologies. No budget for paid APIs, premium hosting, or commercial software licenses.

Hardware Limitations: The application must function on low-end smartphones with limited memory (1-2GB RAM) and older browsers. Page sizes must be optimized to load on 3G networks (target: under 500KB per page).

Language Requirements: The interface must support both English and Kinyarwanda. All text content must be externalized for easy translation. Database must use UTF-8 encoding for Kinyarwanda characters.

Regulatory Compliance: Must comply with Rwanda's Law No. 058/2021 on the Protection of Personal Data and Privacy. Student data must be handled according to data protection principles.

Security Considerations: All communications must use HTTPS/TLS 1.2+. Passwords must be hashed using bcrypt. SQL injection and XSS vulnerabilities must be prevented through parameterized queries and input sanitization.

Timeline Constraint: MVP must be completed within 8 weeks using Agile methodology with 2-week sprints. This limits the feature set to core functionality only.

Hosting Constraint: Preferably hosted on servers located in Rwanda or East Africa to minimize latency and comply with data sovereignty requirements.

2.6 User Documentation

The following documentation will be provided:

(1) In-app help tooltips and guidance text throughout the interface;

(2) A simple FAQ page accessible from the main menu;

(3) A brief user guide (PDF, 2-3 pages) explaining how to register, take the assessment, and interpret results;

(4) Video tutorial (2-3 minutes) demonstrating the assessment process, available in both English and Kinyarwanda.

2.7 Assumptions and Dependencies

Assumptions:

(1) Students have access to internet-enabled devices (smartphones or computers) at school or home;

(2) Schools will allow students to use the platform during career guidance sessions;

(3) Career data for Rwanda will be available from public sources (NISR, RDB, universities).

Dependencies:

(1) Web hosting service availability;

(2) Accurate and up-to-date career and education data from Rwandan institutions;

(3) Cooperation from pilot schools for testing and feedback;

(4) Availability of Kinyarwanda translation resources.

3. External Interface Requirements

3.1 User Interfaces

The user interface will be a responsive web application with the following screens:

(1) Home/Landing Page - welcome message, login/register buttons;

(2) Registration Page - form for email, password, name, school, grade;

(3) Login Page - email and password fields;

(4) Dashboard - overview of assessment status and results;

(5) Assessment Page - questionnaire with progress indicator, one question at a time or grouped;

(6) Results Page - career matches displayed as cards with percentage scores;

(7) Career Detail Page - full information about selected career including education pathways.

Design principles: clean, minimal design; large touch-friendly buttons; high contrast colors; clear fonts (minimum 16px); language toggle (EN/KIN) visible on all pages.

3.2 Hardware Interfaces

The system is a web application and does not directly interface with specific hardware. Users access the system through standard web browsers on their devices (smartphones, tablets, laptops, desktop computers). The server hardware requirements are minimal: a standard cloud server with at least 1GB RAM and 20GB storage is sufficient for the MVP. No specialized hardware interfaces are required.

3.3 Software Interfaces

Database: MySQL 8.0 for storing user data, assessment questions, responses, and career information. Web Server: Apache.

Backend: PHP 8.0+ for server-side logic.

Frontend: JavaScript (lightweight framework).

The system may integrate with email services (SMTP) for password reset functionality. All data exchange between client and server uses JSON format over HTTPS.

3.4 Communications Interfaces

The system uses standard web communication protocols: HTTPS (TLS 1.2+) for all data transmission to ensure security; HTTP/1.1 or HTTP/2 for web requests; RESTful API design for client-server communication. Email communication via SMTP for password reset emails. No real-time communication features (like WebSockets) are required for the MVP. Data transfer is optimized for low-bandwidth connections (compressed responses, minimal image sizes).

4. Requirement Specification

EXAMPLE : (EDIT OUT THIS PART BASED ON YOUR SPECIFIC REQUIREMENTS)

Stakeholder Requirements Specification

Functional Requirements

Req ID	Requirements	Description
FR 1	**User Management**	
FR 1.1	User Registration	Allow students to create accounts using email, name, school name, and grade level (S1-S6)
FR 1.2	User Login	Allow registered users to log in with email and password
FR 2	**Career Assessment**	
FR 2.1	Interest Questionnaire	Present 25-30 questions about student interests, skills, and preferences
FR 2.2	Assessment Scoring	Calculate scores across different career categories based on responses
FR 3	**Career Results**	
FR 3.1	Career Matching	Display top 3-5 career matches based on assessment scores with match percentage
FR 3.2	Career Information	Display detailed career info: description, skills, education path, salary range in Rwanda
FR 4	**Education Info**	
FR 4.1	University Info	Show relevant Rwandan universities and TVET institutions for each career
FR 4.2	Save Results	Save assessment results so students can view them later without retaking
FR 4.3	Retake Assessment	Allow students to retake the assessment to update their results
FR 5	**Language Support**	
FR 5.1	Bilingual Interface	Support both English and Kinyarwanda languages for interface and assessment

5. Other Nonfunctional Requirements

5.1 Example of Non-Functional Requirements (EDIT THIS PART)

Requirement Type	Req ID	Description
Security	NFR 1	User passwords shall be encrypted using secure hashing (bcrypt)
Performance	NFR 2	Pages shall load within 5 seconds on 4G connection. The system supports 50 concurrent users.
Usability	NFR 4	Interface simple enough for students to complete registration and assessment without help
Auditability	NFR 5	System available 95% of the time during school hours (7am-6pm EAT)
Cross Browser Support	NFR 6	The system must be accessible across the two web browser ● Google Chrome ● Microsoft Edge
Technology	NFR 7	● The system must be accessible on: PC (Windows, Mac, Linux), Android smartphones and tablets, iOS devices (iPhone, iPad). ● The system must be accessible via any internet connection (WiFi, 3G, 4G mobile data)

6. Appendix

Additional requirements: Database must support UTF-8 encoding for Kinyarwanda characters. The system should comply with Rwanda's data protection guidelines. All user data must be stored within servers located in Africa (preferably Rwanda or East Africa) for data sovereignty compliance.

Appendix A: Glossary

- MVP: Minimum Viable Product - the simplest version with core features.
- TVET: Technical and Vocational Education and Training.
- S1-S6: Senior 1 through Senior 6 (Rwandan secondary school grades).
- EAT: East Africa Time (UTC+2).
- NISR: National Institute of Statistics of Rwanda.
- REB: Rwanda Education Board.
- HTTPS: Hypertext Transfer Protocol Secure. API: Application Programming Interface.

Appendix B: Analysis Models

Analysis models will be developed during the design phase. Planned models include:

(1) Entity-Relationship Diagram showing User, Assessment, Question, Response, Career, and Institution entities;

(2) User flow diagram showing the student journey from registration to viewing results;

(3) Simple system architecture diagram showing client, server, and database components.